SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451

Internet of Things (IoT) and Applications

IoT Protocol-(Coap-MQTT-DDS-AMQP)

Assoc. Prof. Cüneyt BAYILMIŞ
Researcher Dr. Ünal ÇAVUŞOĞLU

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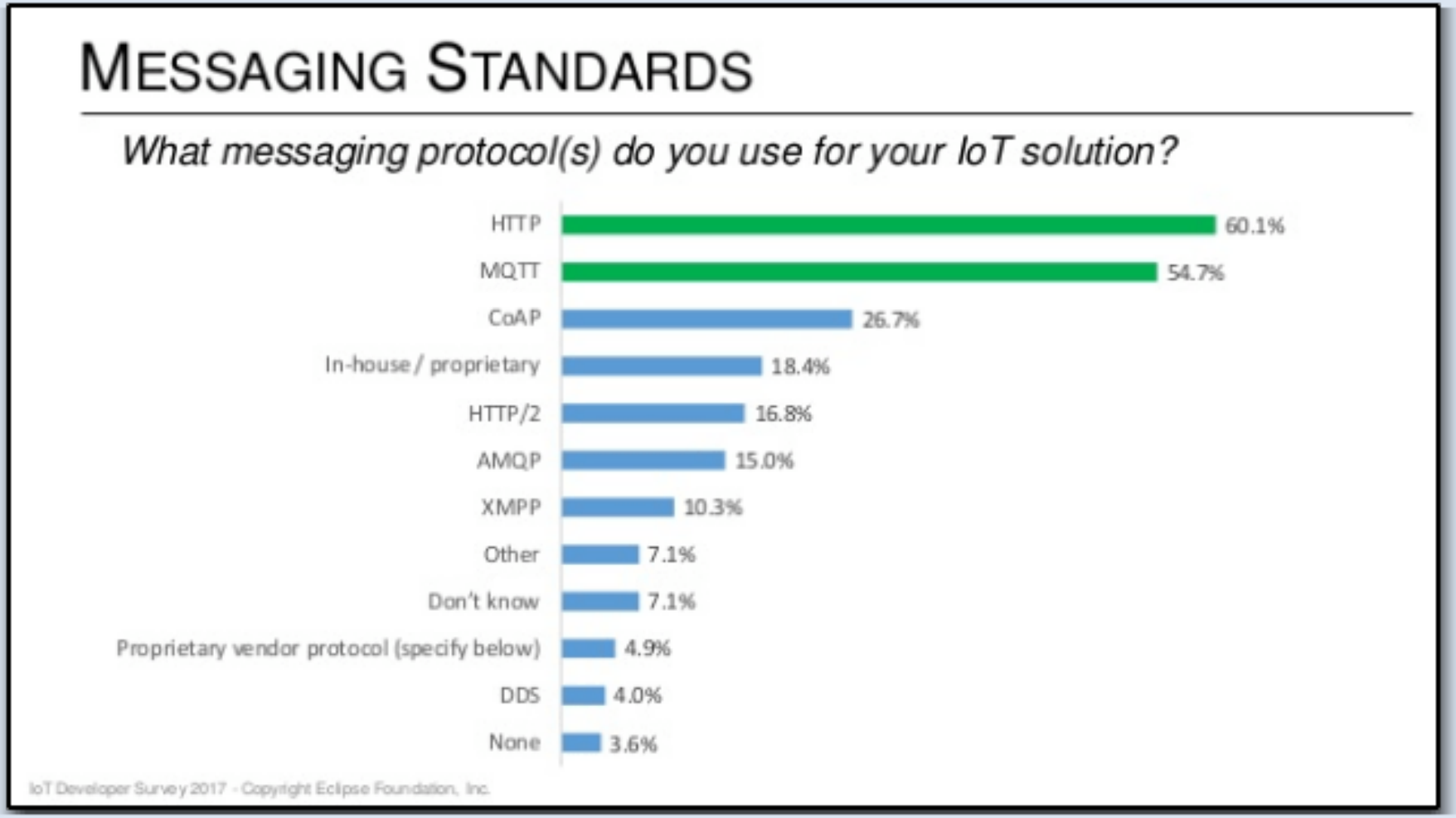
Application Protocols of IoT

- REpresentational State Transfer, (REST)
- Simple Object Access Protocol, (SOAP)
- Extensible Messaging and Presence Protocol, (XMPP)
- Constrained Application Protocol, (CoAP)
- Messega Queue Telemetry Transport, (MQTT)
- Advanced Message Queuing Protocol, (AMQP)
- Data Distribution Service, (DDS)
- OMA LightwieghtM2M (LWM2M)
- oneM2M

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Application Protocols of IoT



Constrained Application Protocol, (CoAP)

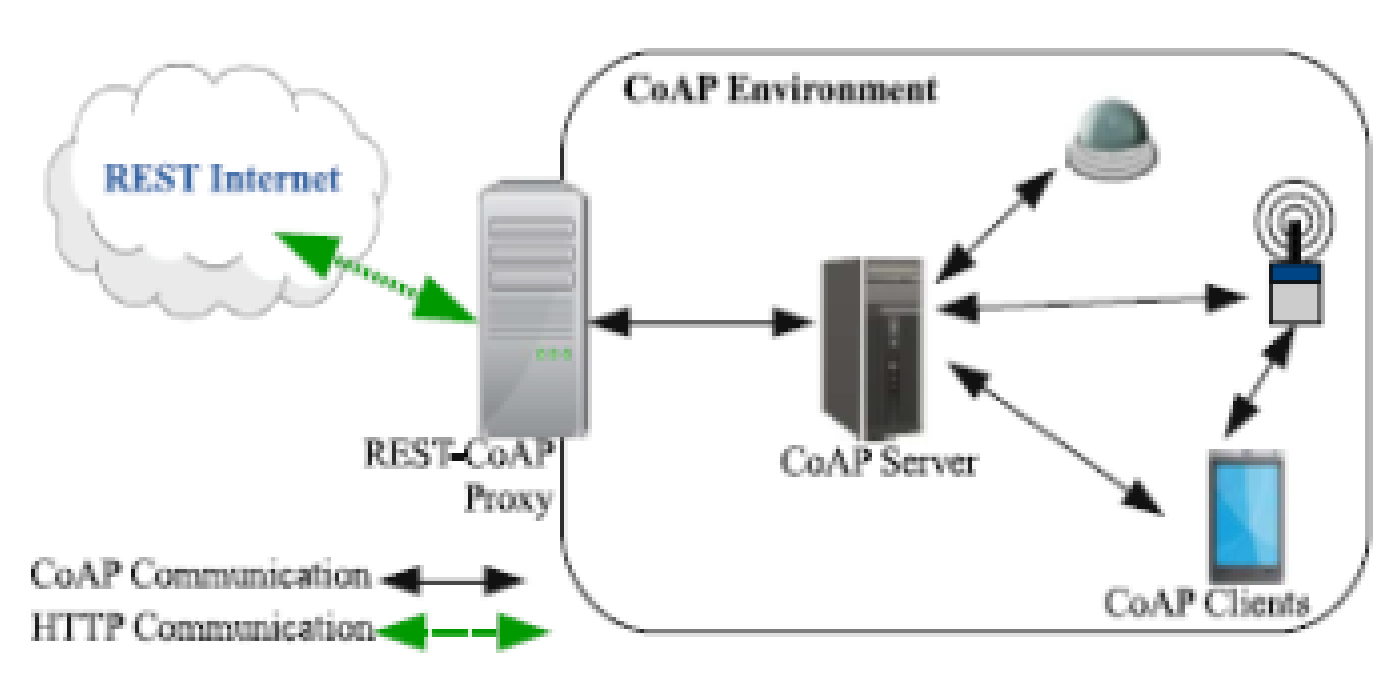
(Constrained Application Protocol, CoAP)

- CoAP was co-implemented by the Internet Engineering Task Force (IETF) and ARM.
- It was standardized as RFC7252 in June 2014.
- CoAP is a customized web transfer protocol designed for applications of internet (inter-machine communication) of objects, limited / restricted nodes such as power, processing capacity, memory and networks.
- CoAP is a web transfer protocol based on Representational State Transfer (REST) above HTTP functions.
- CoAP is based on client / server model using request / response communication.

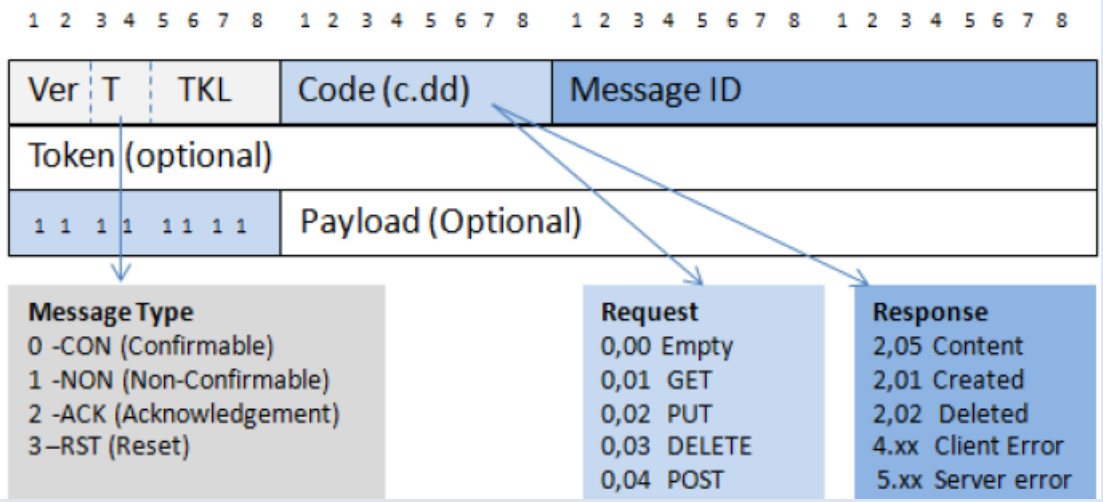
  GET, PUT, POST, and DELETE methods are used in HTTP.
- CoAP is a lightweight protocol used over UDP.
  - CoAP packets are much smaller than HTTP TCP streams. UDP uses only 8 bytes as the packet header. 4 bytes are appended by the CoAP as the application title.
  - Clients and servers communicate via disconnected datagrams. Since it is a datagram-based protocol, it can be used in other packet-based communication protocols.
  - CoAP allows UDP broadcast and multicast mechanisms for addressing.

(Constrained Application Protocol, CoAP)

- CoAP supports publisher / subscriber architecture due to multicast communication support. CoAP with asynchronous message exchanges is customized for IOT.
- CoAP secure communication is provided by using Datagram Transport Layer Security (DTLS) or Transport Layer Security (TLS) like datagrams.
- CoAP Structure

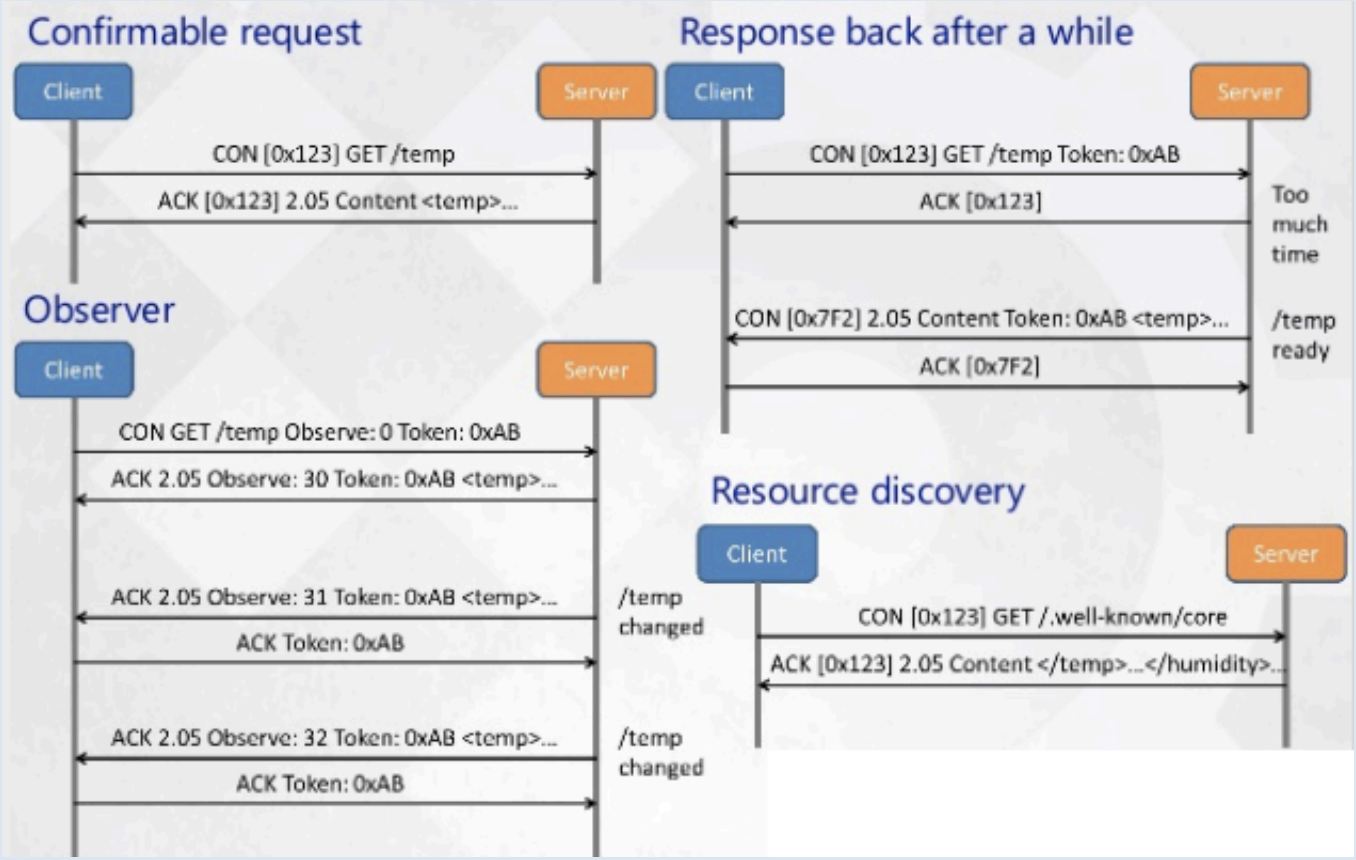


CoAP Message Structure



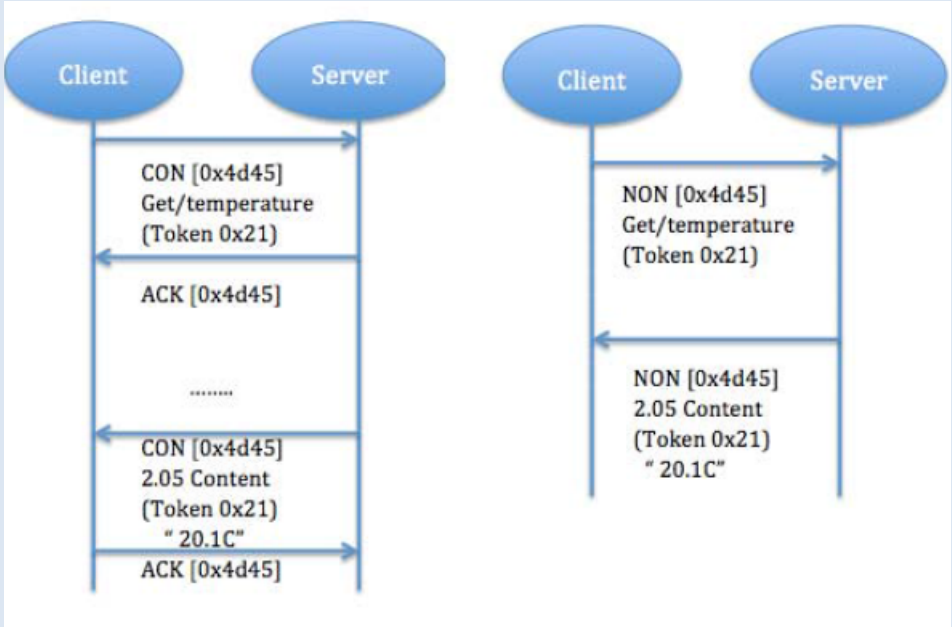
- Ver: CoAP protocol version (2 bits),
- T: Transaction Type (2 bits)
- TKL (Token Length): Specifies the size of the Token between 0 and 8 bytes.
- The code shows the results of the communication between the client and the server.
- Message ID: Used as a message identifier to provide secure communication and to detect message copies.
- Token: Requests and Answers address each other (with token-id).
- Payload: Load

CoAP Request / Response Protocol Example



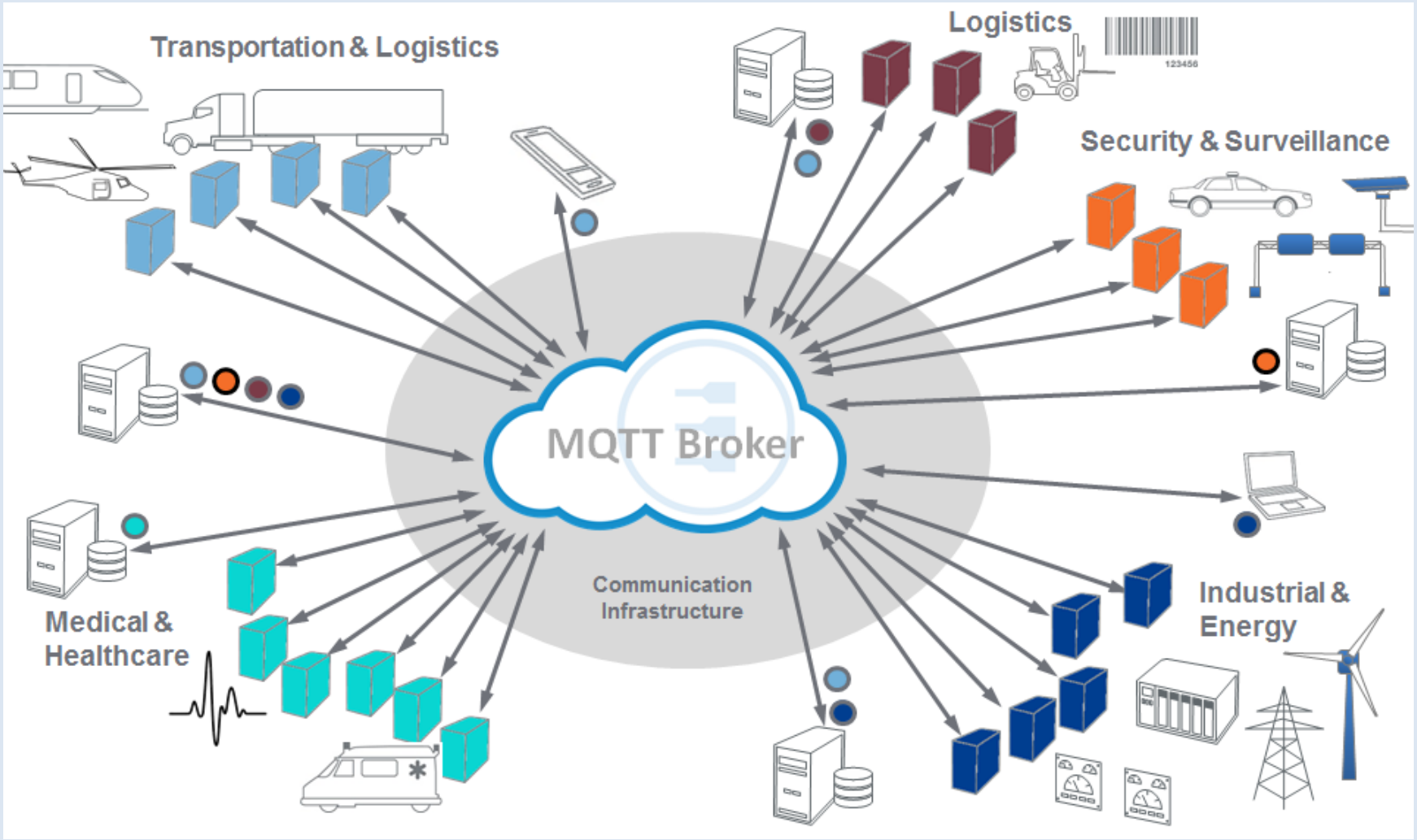
- When Confirmable (CON) communication is selected, an ACK packet is received / sent that processes are completed (messages are received).
- When non-Confirmable (NON) communication is selected, ACK is not used.
- The correspondence between Request and Response is made by Message ID and Token-ID.

CoAP Request / Response Protocol Example



- When Confirmable (CON) communication is selected, an ACK packet is received / sent that processes are completed (messages are received).
- When non-Confirmable (NON) communication is selected, ACK is not used.
- The correspondence between Request and Response is made with Message ID (0x4D45) and Token-ID (0x21).

(Messega Queue Telemetry Transport, MQTT)

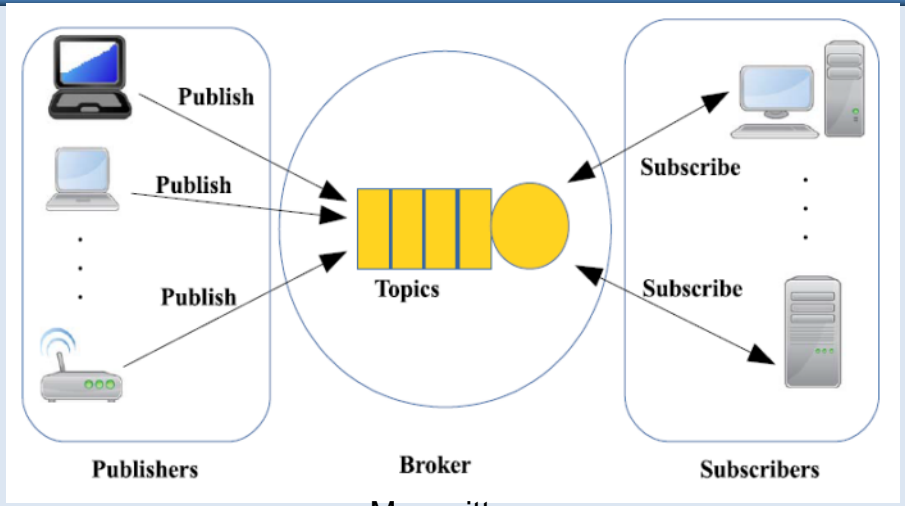


Messega Queue Telemetry Transport (MQTT)

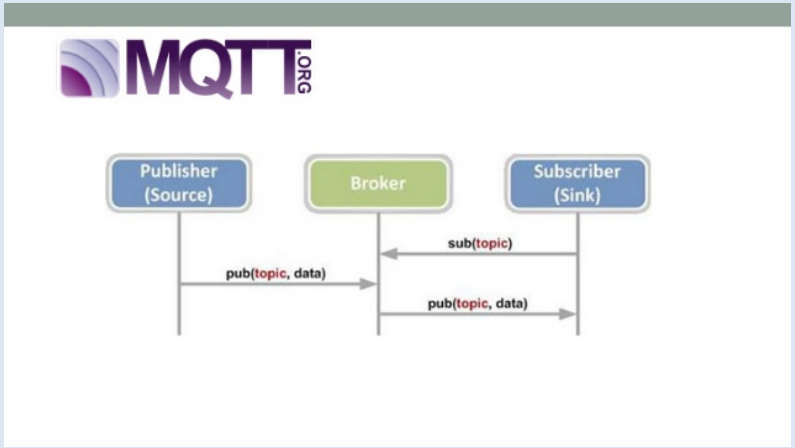
- MQTT is a messaging protocol used to communicate objects between the Internet or between machines.
- It was introduced by IBM and Arcom (today Eurotech) in 1999 but standardized OASIS in 2013.
- MQTT aims to provide embedded systems and network (cloud) connectivity.
- In other words, it allows you to control and monitor networks of small devices over the cloud.
- MQTT is a telemetry messaging protocol based on the ability to be publisher and subscriber.
- Highlights of the MQTT protocol are the ability to store messages and resend them when a connection is established.
- MQTT is based on the TCP protocol.
- MQTT messages are transmitted according to 3 different quality of service (QoS) levels. Best Effort, At Least Once, Exactly Once
- There are two types of MQTT: MQTT v3.1 and MQTT-SN (MQTT for Sensor Network)

Messega Queue Telemetry Transport (MQTT)

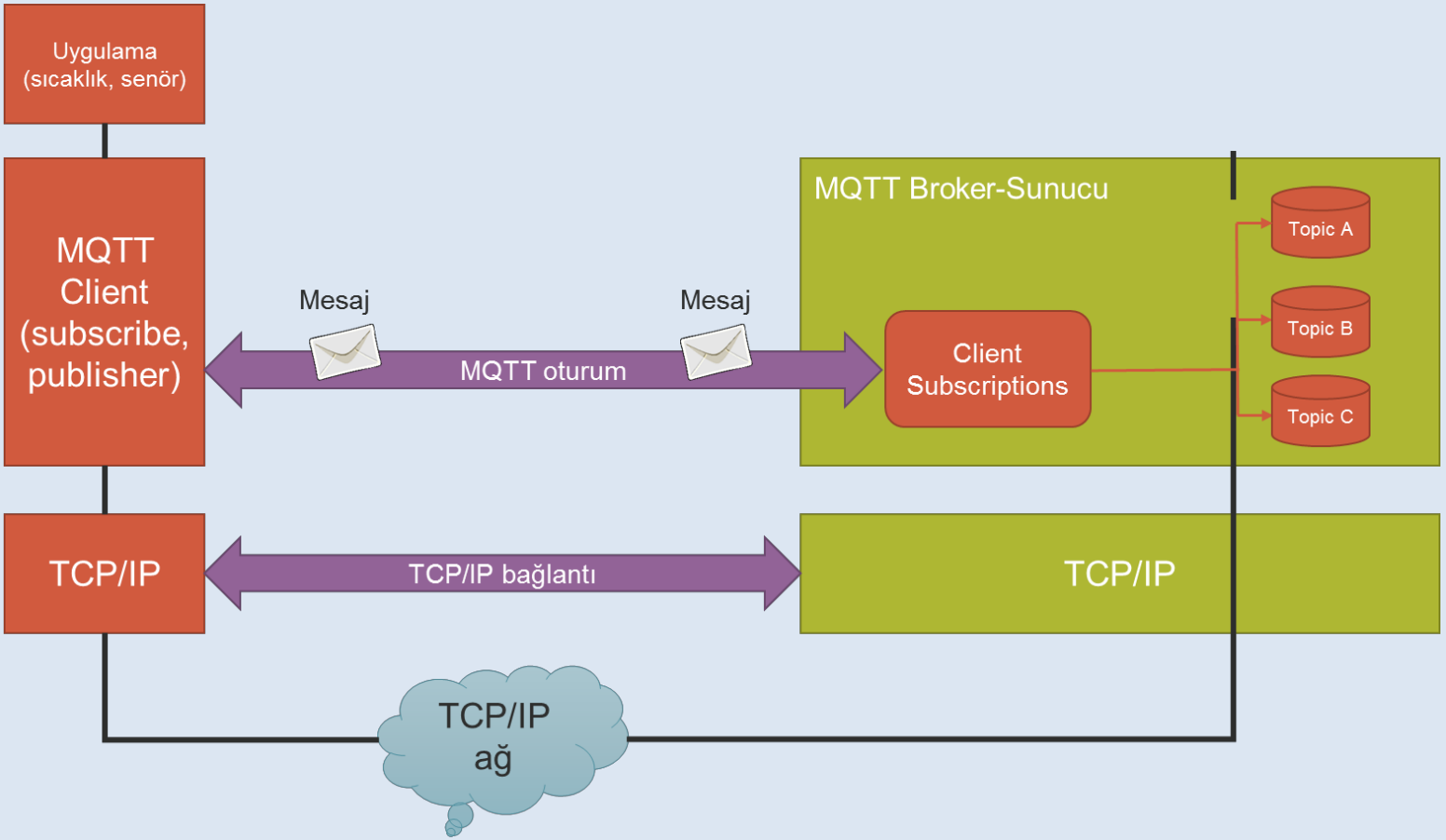
- MQTT, subscriber, Publisher, and server (broker).
- Devices in the MQTT architecture can publish messages using an MQTT broker, or register to listen to messages.
- The message consists of two main parts: topic and payload.
- Publisher publishes messages on a specific topic, only those who subscribe to the message.
- For example, a sensor periodically broadcasts measurement values, and subscribers to this sensor receive this message.



Mosquitto
HiveMQ
Mosca
Paho

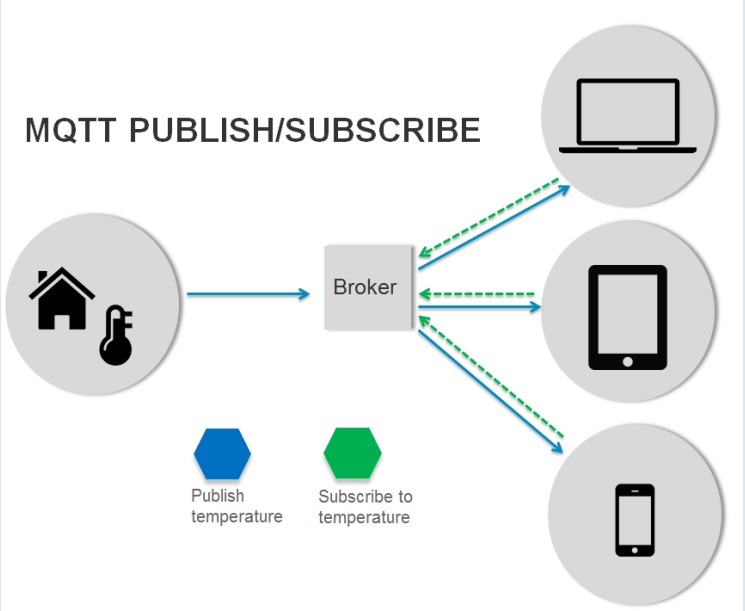


MQTT Katmanlı Mimarisi



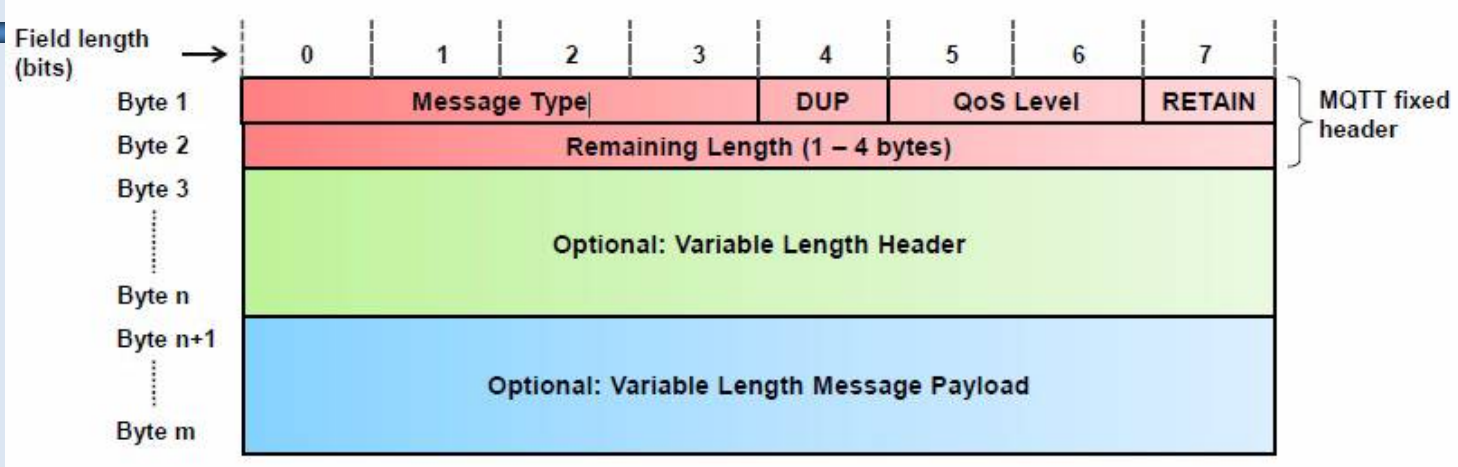
MQTT Broker

- M2M devices can publish messages with MQTT brokers, or they can subscribe to listen to messages.
- A broker is a program that performs the task of a TCP server and has a dynamic database.
- Receive packages are kept in the database indexed according to topics.
- If there is no subscriber for a package, that package is discarded from the database.
- If the subscriber is not in contact, the packet is deleted after the specified time.
- With the Topic feature, multiple, multiple, many, many links can be established in the MQTT protocol.
- The default port for MQTT Broker is 1883.



- Mosquitto: One of the most common MQTT brokers. It has been developed in C language and is a platform for M2M communication according to MQTT protocol standards.

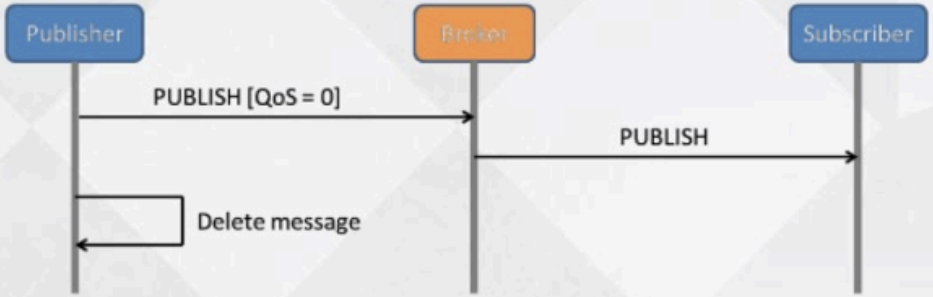
MQTT Message Structure



- The minimum message size for MQTT is 2 bytes. (C # M2Mqtt library 30 KB)
- The Message Type field shows the message CONNECT, CONNACK, PUBLISH, SUBSCRIBE.
- The DUP is a flag that duplicates the message.
- The QoS Level shows the QoS level at which the transmitted message will be transmitted. QoS 0, message may not be transmitted, minimum traffic. QoS 1, the sent message is stored in its own database and can be sent again and again. The message is transmitted strictly, can be transmitted more than once. QoS 2 is similar to QoS 1, but a response information is received when message transmission is performed. The message is transmitted in one pass and precisely. Maximum traffic.
- Retain notifies the server of the last received publish message and delivers it as the first message to the new subscriber.
- Remaining Length indicates the remaining size of the message.
- Header part, protocol version etc. information.
- The payload section contains information such as topic, message, user, and password.

MQTT Quality of Services (QoS)

QoS 0 : At most once (fire and forget)



QoS 1 : At least once



QoS 2 : Exactly once



The security in MQTT Protocol

- MQTT, based on the TCP protocol, uses SSL / TLS for security.
- The connection uses the username / password in the CONNECT message.
- The load on the MQTT message structure is transmitted encrypted.

Messega Queue Telemetry Transport (MQTT)

- Many applications use MQTT such as health care, energy meter, monitoring, Facebook notification.
- MQTT, general characteristics

✓ Asynchronous protocol,

✓ Provides one to one, one to many, many to many connections.

✓ It has a broker-based communication mechanism.

✓ Topic based addressing.

✓ It supports SSL / TLS as security.

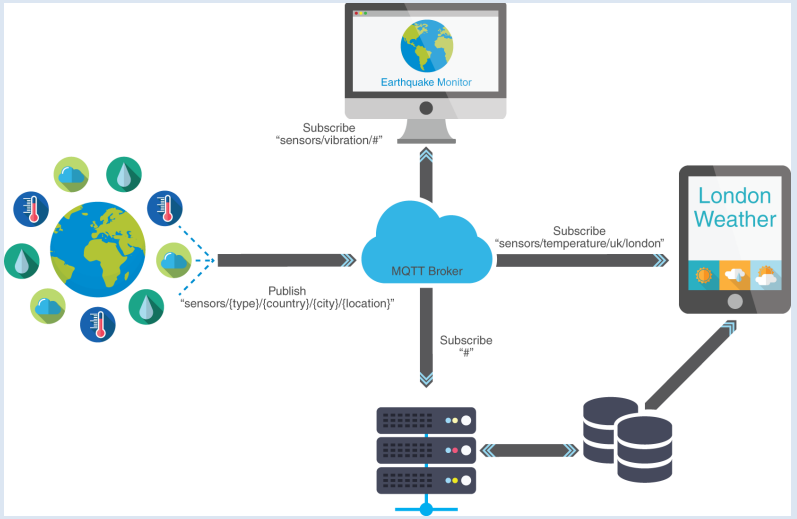
✓ Minimizes resource use.

✓ It provides a communication at the millisecond level.

✓ It uses the TCP / IP connection type.

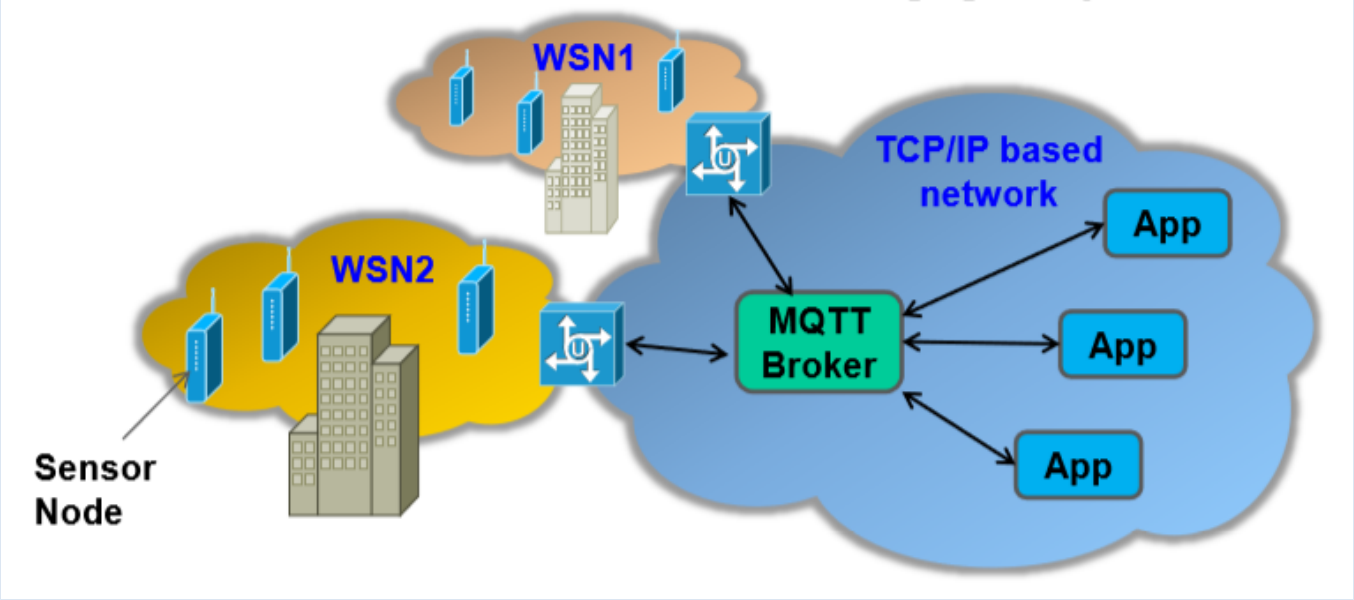
✓ The default port is 1883.

✓ TCP / IP protocol can be written for Linux, Windows, works on Android, iOS, MacOS.



MQTT-SN

- It is the MQTT protocol version for wireless sensor networks.
- MQTT-S does not use the TCP / IP protocol on the WSN transport layer. It has its own protocol stacks. For this reason, it does not work directly on TCP / IP.
- WSN is connected to traditional TCP / IP network with the help of gateway devices.
- Some of the changes in MQTT-S include replacing Topic by ID, for example the lack of registration requirement for predefined Topic IDs.



Why MQTT?

- Energy consumption is lower than HTTP protocol.
- Comparison of sustained link cost

	% Battery / Hour			
	3G		Wifi	
Keep Alive (Seconds)	HTTPS	MQTT	HTTPS	MQTT
60	1.11553	**0.72465**	0.15839	**0.01055**
120	0.48697	**0.32041**	0.08774	**0.00478**
240	0.33277	**0.16027**	0.02897	**0.00230**
480	0.08263	**0.07991**	0.00824	**0.00112**

- Receive (1024 message / 1 bayt)

	3G		Wifi	
	HTTPS	MQTT	HTTPS	MQTT
% Battery / Hour	18.43%	**16.13%**	**3.45%**	4.23%
Messages / Hour	1708	**160278**	3628	**263314**
% Battery / Message *	0.01709	**0.00010**	0.00095	**0.00002**
Messages Received	240 / 1024	**1024 / 1024**	524 / 1024	**1024 / 1024**

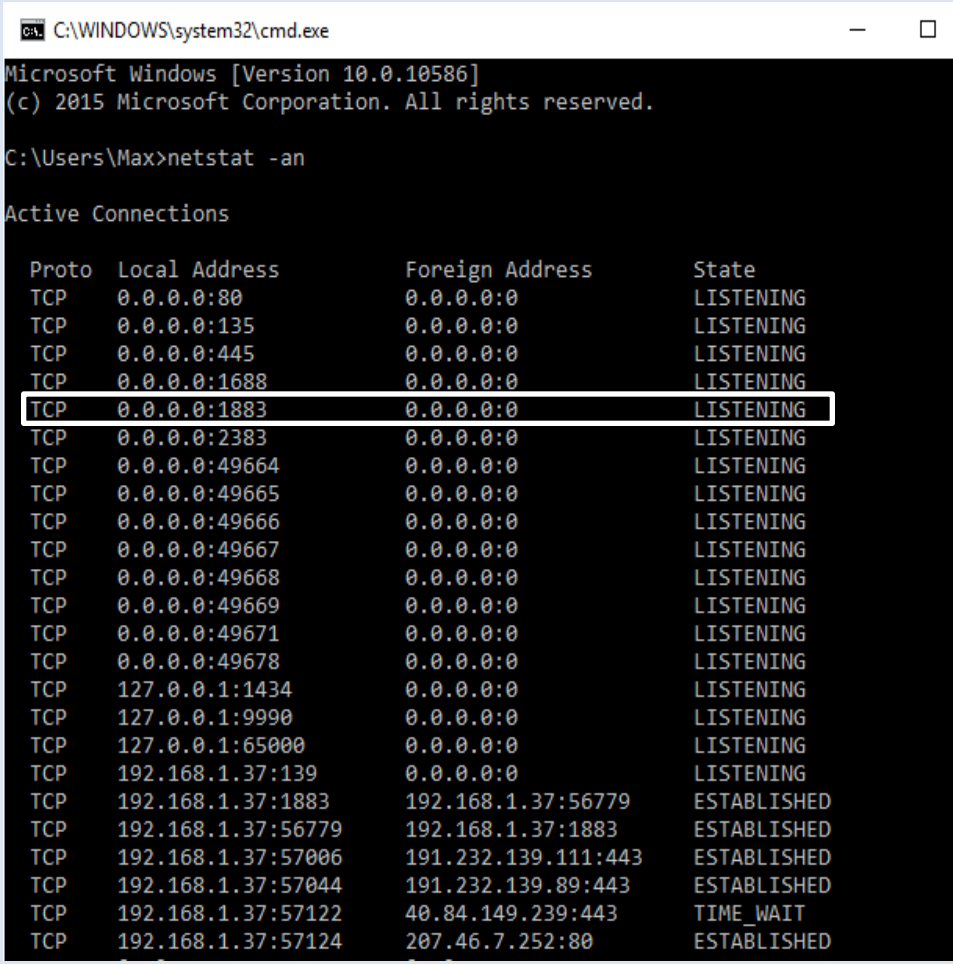
- Send (1024 message / 1 bayt)

	3G		Wifi	
	HTTPS	MQTT	HTTPS	MQTT
% Battery / Hour	18.79%	**17.80%**	5.44%	**3.66%**
Messages / Hour	1926	**21685**	5229	**23184**
% Battery / Message *	0.00975	**0.00082**	0.00104	**0.00016**

Kaynak: http://stephendnicholas.com/posts/power-profiling-mqtt-vs-https

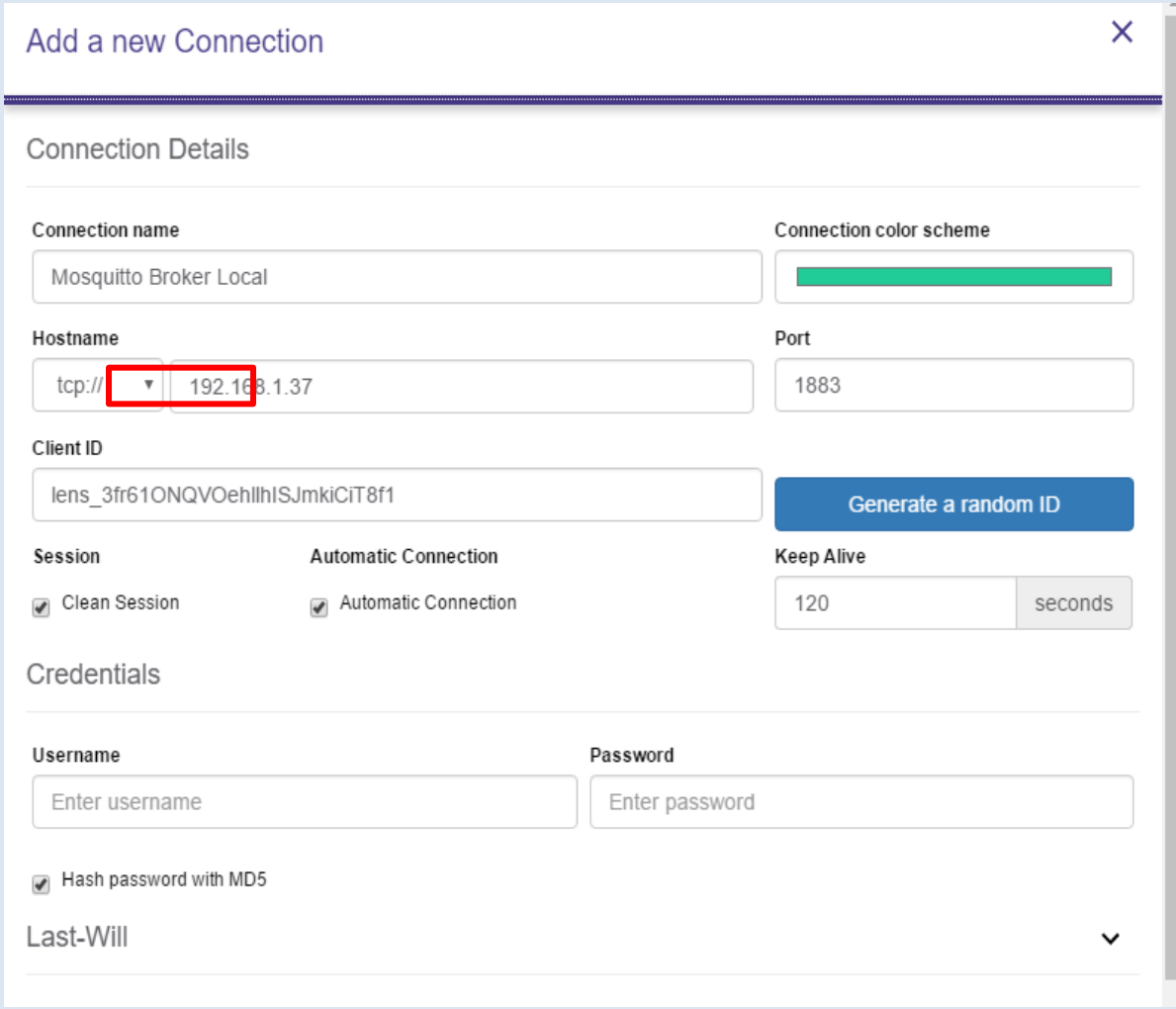
Mosquitto MQTT Broker Installation in Windows

- ✓ Download mosquitto from the official site http://mosquitto.org/js/mosquitto-1.1.js and install it.
- ✓ We are installing Win32 OpenSSL.
- ✓ Paste the files libeary32.dll, ssleay32.dll, pthreadVC2.dll into the C: \ Windows \ System32 folder.
- ✓ To check if Mosquitto is running, type netstat -an at the command line and check that port 1883 is open.

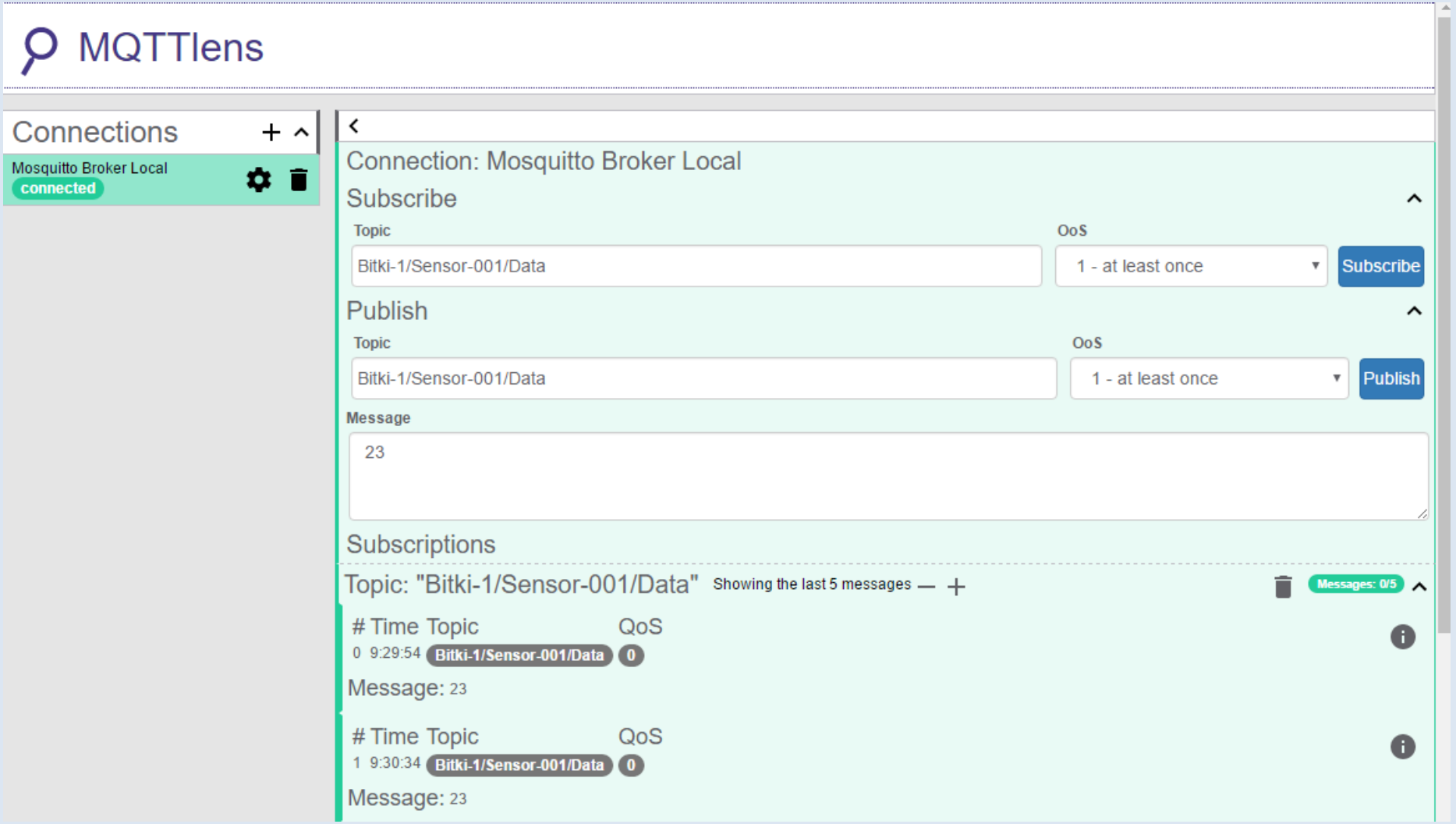


MQTT Broker Local Connection (MQTT lens)

- MQTT lens application is a Chrome browser plug-in that communicates with the MQTT Broker.
- With this add-on, the communication between Publisher and subscriber of MQTT messages can be tested.
- In the hostname box, enter the local ip address and press the connection button.

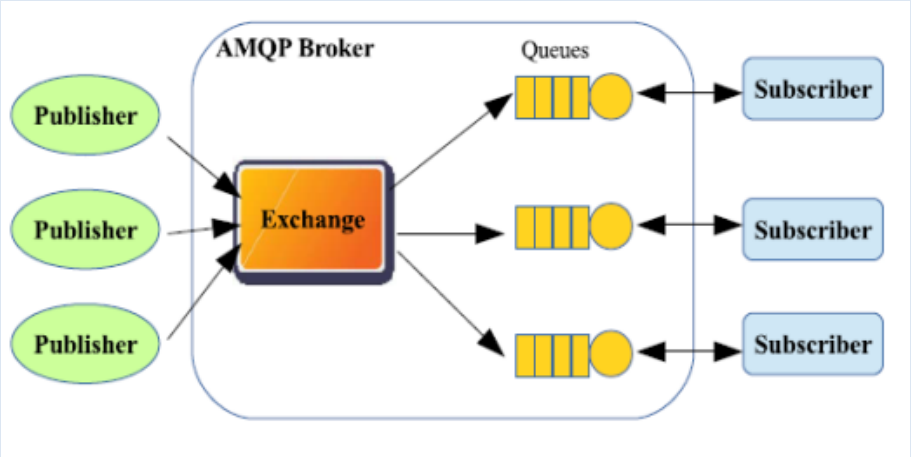


MQTT Broker Local Connection (MQTT lens)



(Advanced Message Queuing Protocol, AMQP)

- AMQP is the open standard application protocol for IOT, which focuses on message routing.
- It is a protocol that provides communications with large data structures, providing servers with queues (brokers) and publishers (publishers) and subscriptions.
- It provides reliable communication with message delivery guarantee at 3 different levels (at-most-once / best effort, at-least-once, exactly once delivery).
- AMQP is provided with communication exchanges and message queues. Exchanges forward messages to the appropriate queue. This redirection is based on predefined rules and bindings.
- In addition to point-to-point communication, AMQP also supports the publish / subscribe communication model.
- AMQP Structure

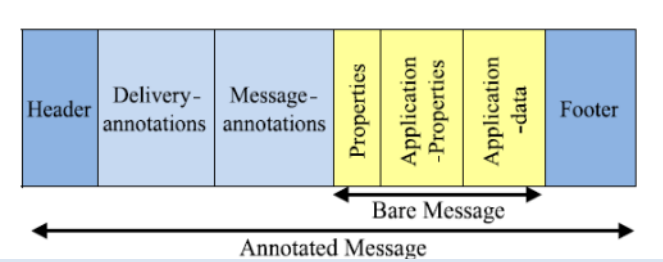


(Advanced Message Queuing Protocol, AMQP)

- AMQP uses TCP as the transport layer. It defines the layer of messaging over this layer.
- AMQP uses two types of messages.

  Message: Messages from the sender.

  Annotated message: The message that appears on the recipient.
- AMQP communication takes place in the millennium.
- The minimum packet size is 60 bytes.
- It is an asynchronous protocol like MQTT.



(Data Distribution Service, DDS)

- DDS was developed by the Object Management Group (OMG) for real-time M2M communications.
- DDS offers advantages such as fewer layers, less workload.
- DDS is the publish-subscribe protocol.
- It supports real-time communication with micro-cellular level communication.
- MQTT in Publish-Subscribe, and AMQP protocol offer more QoS support than others. It provides 23 QoS level / policy considering communication criteria such as secure, urgent / urgent, priority, reliable.
- It supports both UDP and TCP as the transport layer.
- DDS has a two-tiered architecture.

Data-Centric Publish-Subscribe (DCPS): Responsible for submitting information to subscriptions.

Data-Local Reconstruction Layer (DLRL): Provides an optional layer and provides DCPS functions. It makes it easier to share distributed data among distributed objects.

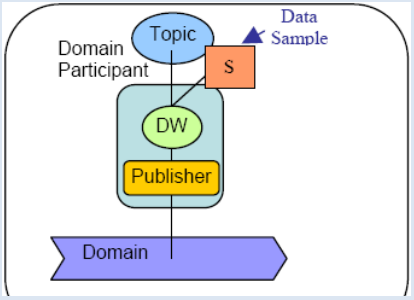
- It supports Datagram TLS (DTLS) with SSL / TLS as security.
- Content-aware routing (payload review).

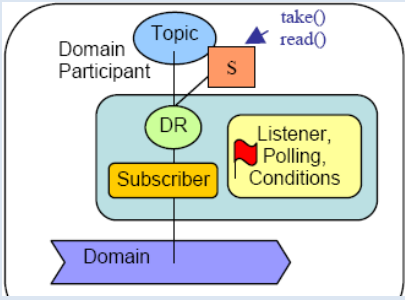
(Data Distribution Service, DDS)

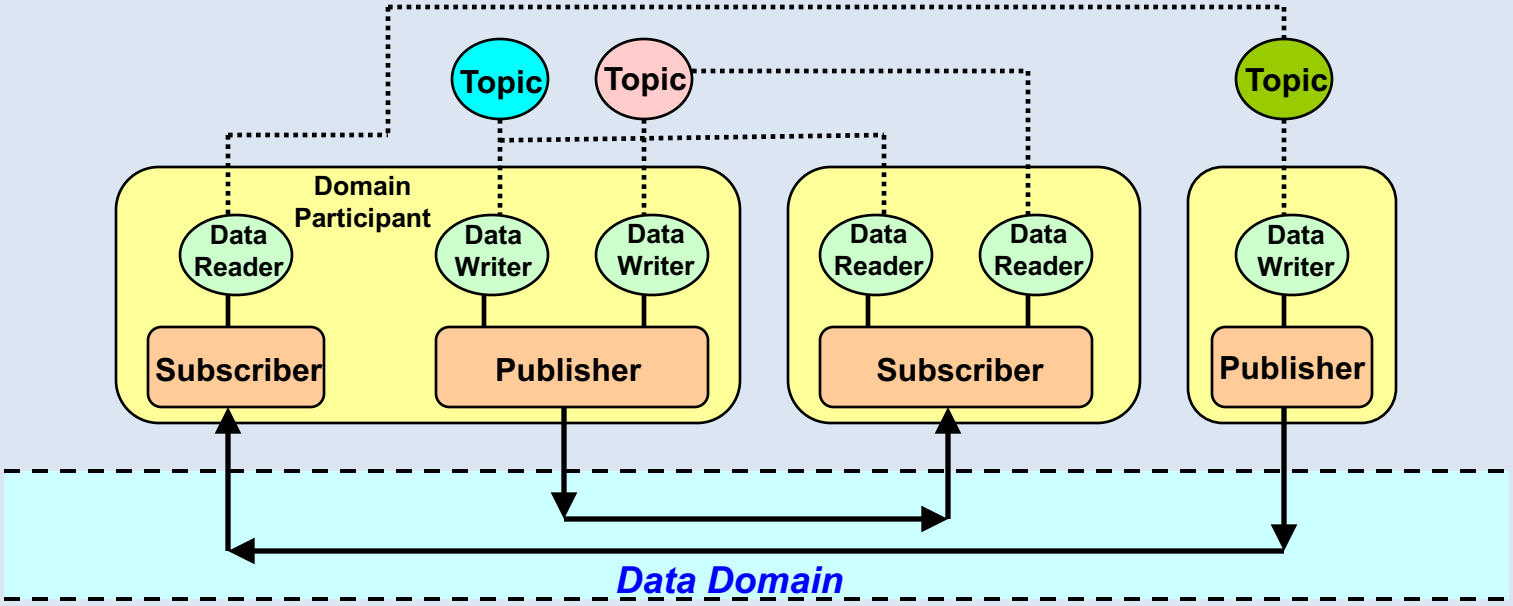
- Topics: Data distribution issues
- Data Writers: Data producers.

  It creates the sender with Publisher.
- Data Readers: Data consumers / receivers.

  It creates the receiver together with the Subscriber.
- Topics are configured with Data Readers / Writers QoS policies.





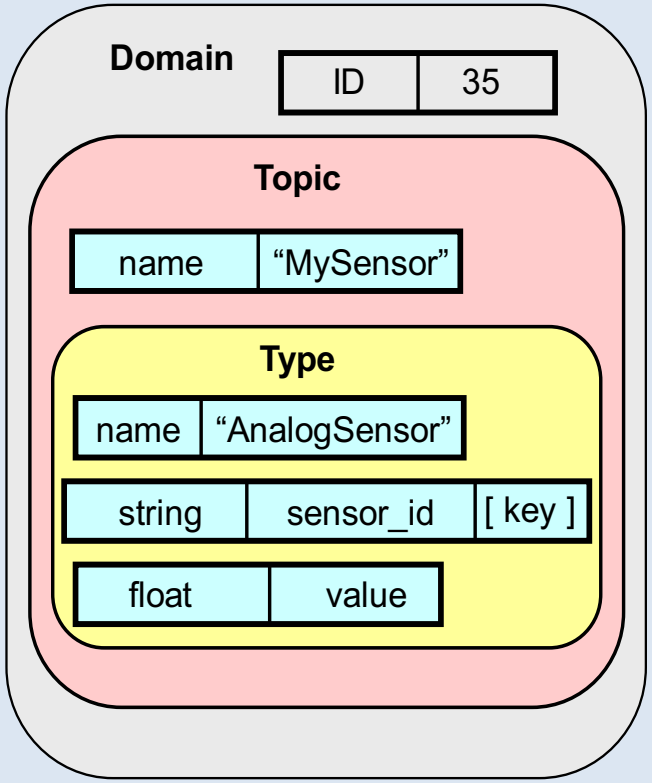


(Data Distribution Service, DDS)

- Topics is the link between Publisher and the subscriber.
- Topic, Name, and Type fields.

  Name is unique in the domain.

  More than one topic can have the same type.
- For content aware subscriptions, the constructs such as multitopics and content-filtered topics are provided

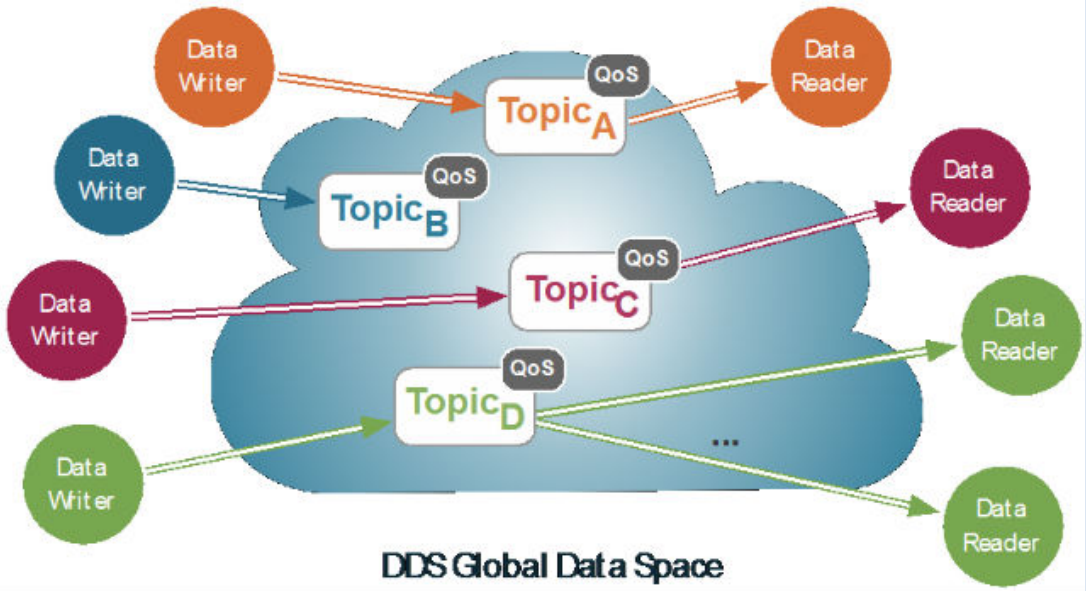


(Data Distribution Service, DDS)

- DDS QoS Protokols
  - Deadline
  - Destination Order
  - Durability
  - Entity Factory
  - Group Data
  - History
  - Latency Budget
  - Lifespan
  - Liveliness
  - Ownership
  - Ownership Strength
  - Partition
  - Presentation
  - Reader Data Lifecycle
  - Reliability
  - Resource Limits
  - Time-Based Filter
  - Topic Data
  - Transport Priority
  - User Data
  - Writer Data Lifecycle

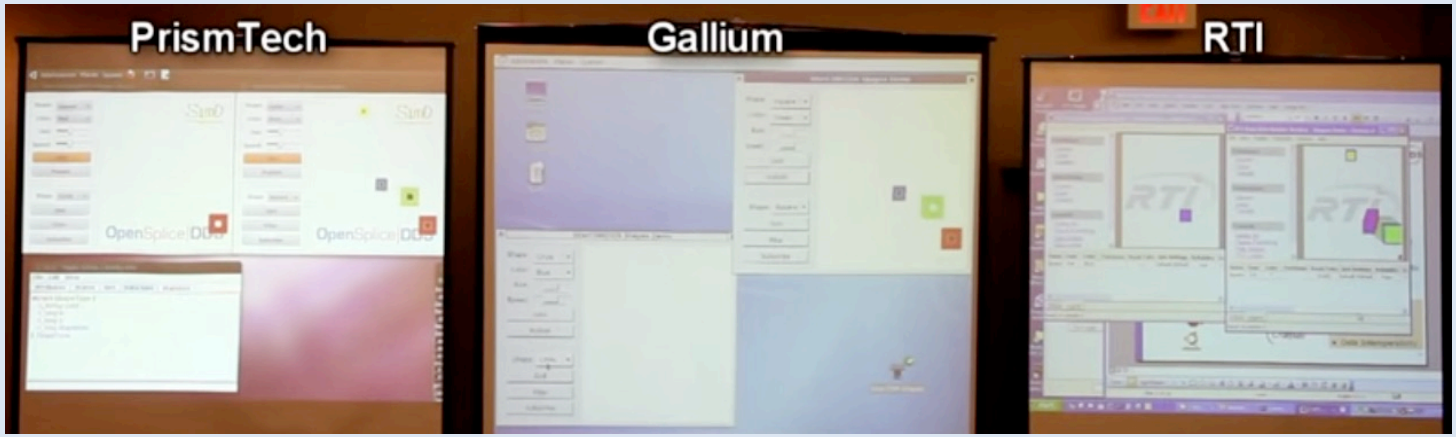
(Data Distribution Service, DDS)

- Some of the quality of service (QoS) policies supported in the DDS protocol are below.
- ✓ Latency_Budget: Defines end-to-end latency.
- ✓ Time_Based_Filter: Interchange between slow consumers and fast producers.
- ✓ Resource_Limits: Controls resources used by the service.
- ✓ Reliability: Provides real-time delivery of the certificate.
- ✓ History: Controls the content (values) of the retrieved data.



(Data Distribution Service, DDS)

- DDS was first used to destroy enemy / vehicle / weapons location information in the US to destroy the target.
- DDS is used on the Tokyo metropolitan expressway following the vehicle.
- ✓ Usage areas
- ✓ In Smart Grid administrations like Smart City,
- ✓ In smartphone operating systems,
- ✓ In Vehicle and Transportation Systems,
- ✓ In Internet radios,
- ✓ In health care
- DDS Platforms
  - RTI,
  - PrismTech,
  - Gallium,



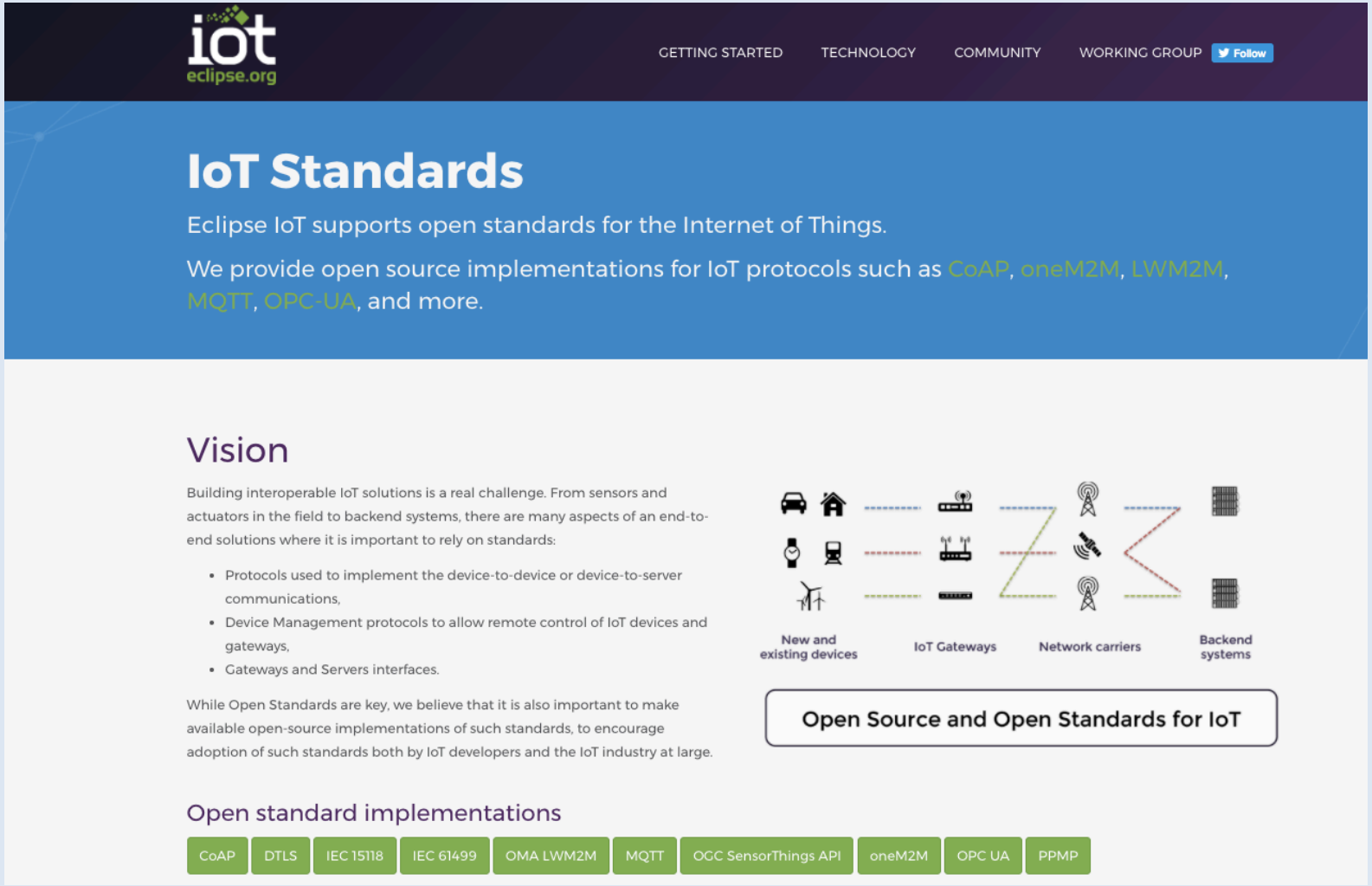
Comparison of IoT Application Protocols

- The most commonly defined protocols of organizations such as IEEE, ETSI.

Application Protocol	RESTful	Transport	Publish/ Subscribe	Request/ Response	Security	QoS	Header Size (Byte)
COAP	✓	UDP	✓	✓	DTLS	✓	4
MQTT	×	TCP	✓	×	SSL	✓	2
MQTT-SN	×	TCP	✓	×	SSL	✓	2
XMPP	×	TCP	✓	✓	SSL	×	-
AMQP	×	TCP	✓	×	SSL	✓	8
DDS	×	TCP UDP	✓	×	SSL DTLS	✓	-
HTTP	✓	TCP	×	✓	SSL	×	-

Implementation of IoT Communication Protocols

- The codes of many IOT Communication Protocols can be accessed at iot.eclipse.org.



References

- A. Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, M. Ayyash, "*Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications*", IEEE Communication Survey&Tutorials, vol. 17 (4), 2347-2376 ,2015.
- L. Atzori, A. Iera, G. Morabito, "The Internet of Things: A Survey", Computer Networks, vol. 54, 2787-2805, 2010.
- O. Vermesan, P. Friess (Editors), "Internet of Things – From Research and Innovation to Market Deployment", River Publishers, 2014.
- M. H. Amaran, N. A. M. Noh, M. S. Rohmad, H. Hashim, "A Comparison of Lightwight Communication Protocols in Robotic Applications", IEEE International Symposium on Robotics and Intelligent Sensors (IRIS2015), 400 – 405, 2015.
- P. Penial, M. Franekova, 'Model of Integration of Embedded Systems via CoAP Protocol of Internet of Things', IEEE International Conference on Applied Electronics, 2016.
- www.mqtt.org
- https://www.oasis-open.org/committees/tc_cat.php?cat=iot
- DDS Uygulama Örneği: https://www.rti.com/downloads/shapes-demo.html
- www.omg.org
- iot.eclipse.org